Lab 7 – Thevenin Parameters (Measured and Simulated)

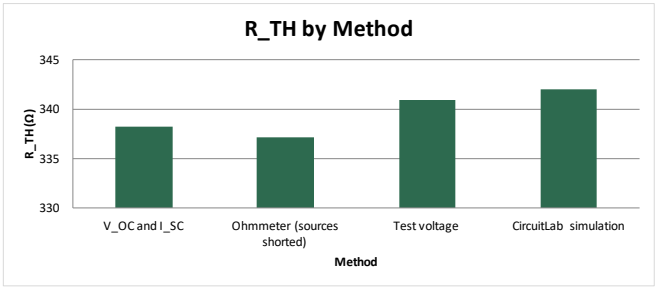
Blue = measured/simulated input values (from lab worksheet & CircuitLab). Black = formulas.

Quantity	Value	Unit	Notes
V_OC (measured)	10.050	V	Step 2, open-circuit across A–B
I_SC (measured)	29.720	mA	Step 2, short-circuit A–B
R_TH = V_OC / I_SC	338.156	Ω	Step 3
R_TH (ohmmeter, sources shorted)	337.200	Ω	Step 5
V_TEST (measured)	10.002	V	Step 7
I_MEASURED	29.336	mA	Step 7
R_TH = V_TEST / I_MEASURED	340.946	Ω	Step 8
V_OC (CircuitLab, corrected)	9.922	V	Step 17, original circuit (corrected model)
I_SC (CircuitLab, corrected)	29.010	mA	Step 17, original circuit (corrected model)
R_TH (CircuitLab) = V_OC / I_SC	342.020	Ω	Step 17

Comparison of Methods (Step 9 / Step 18)

Method	V_TH (V)	R_TH (Ω)	% Diff R_TH vs. Simulation	% Diff R_TH vs. Mean of Bench Methods
V_OC and I_SC	10.050	338.16	-1.13%	-0.18%
Ohmmeter (sources shorted)	10.050	337.20	-1.41%	-0.46%
Test voltage	10.050	340.95	-0.31%	0.64%
CircuitLab simulation	9.922	342.02	0.00%	0.96%
Mean of bench methods	10.050	338.77		
Std. deviation of bench R_TH		1.95		

Note: Methods 2 and 3 do not produce V_TH on their own; V_TH for those rows is the measured open-circuit voltage.



Power Delivered to Load – Original Circuit (Steps 11–13)

P is recomputed as V_L × I_L. Effective R_L shows the actual resistance of the physical load used.

Step	Nominal R_L (Ω)	V_L (V)	I_L (mA)	Effective R_L = V/I (Ω)	P_L = V·I (mW)	P theory (mW)	% Diff vs. Theory
12	100	2.2544	22.754	99.08	51.30	52.35	-2.00%
13	338.156	5.0570	14.555	347.44	73.60	74.66	-1.41%
11	820	7.0878	8.638	820.51	61.23	61.73	-0.82%

Max theoretical power $P_{max} = V_{TH}^2 / (4 \cdot R_{TH})$ (mW): 74.67

Power Delivered to Load – Thevenin Equivalent Circuit (Step 16)

Thevenin equivalent built from a DC supply and series resistors approximating R_TH.

Step	Nominal R_L (Ω)	V_L (V)	I_L (mA)	Effective R_L = V/I (Ω)	P_L = V·I (mW)	P theory (mW)	% Diff vs. Theory
16	100	2.2422	22.543	99.46	50.55	52.46	-3.64%
16	338.156	5.0815	14.866	341.82	75.54	74.67	1.17%
16	820	7.1654	8.723	821.46	62.50	61.70	1.30%

Max theoretical power P_max = V_{TH}^2 / (4·R_TH) (mW): 74.67

CircuitLab Simulation – Thevenin Equivalent (V_TH = 10 V, R_TH = 338.2 Ω)

Simulated load voltage read from CircuitLab voltmeter; current and power computed.

V_TH (sim) 10 R_TH (sim) 338.2

R_L (Ω)	V_L (V)	I_L (mA)	P_L (mW)	P theory (mW)
100	2.282	22.82	52.08	52.08
338.2	5.000	14.78	73.92	73.92
820	7.080	8.63	61.13	61.13

Theoretical Power Transfer Curve vs. Measured Points

P(R_L) = V_TH²·R_L/(R_TH+R_L)², using bench V_OC and R_TH = V_OC/I_SC

R_L (Ω)	P theory (mW)
25	19.15
50	33.52
75	44.38
100	52.61
125	58.86
150	63.58
175	67.12
200	69.75
225	71.66
250	72.99
275	73.88
300	74.40
325	74.64
350	74.65
375	74.47
400	74.15
425	73.70
450	73.17
475	72.56
500	71.89
525	71.17
550	70.42
575	69.65
600	68.85
625	68.05
650	67.23
675	66.42
700	65.60
725	64.79
750	63.98
775	63.17
800	62.38
825	61.59
850	60.81
875	60.05
900	59.30
925	58.55
950	57.83
975	57.11
1000	56.41
1025	55.71
1050	55.04
1075	54.37
1100	53.72
1125	53.08
1150	52.45
1175	51.83
1200	51.23
1225	50.64
1250	50.06
1275	49.49
1300	48.93
1325	48.38
1350	47.85
1375	47.32
1400	46.80
1425	46.30
1450	45.80
1475	45.32
1500	44.84

R_L orig (Ω)	P orig (mW)	R_L Thev (Ω)	P Thev (mW)	R_L sim (Ω)	P sim (mW)
99.08	51.30	99.46	50.55	100.00	52.08
347.44	73.60	341.82	75.54	338.20	73.92
820.51	61.23	821.46	62.50	820.00	61.13

